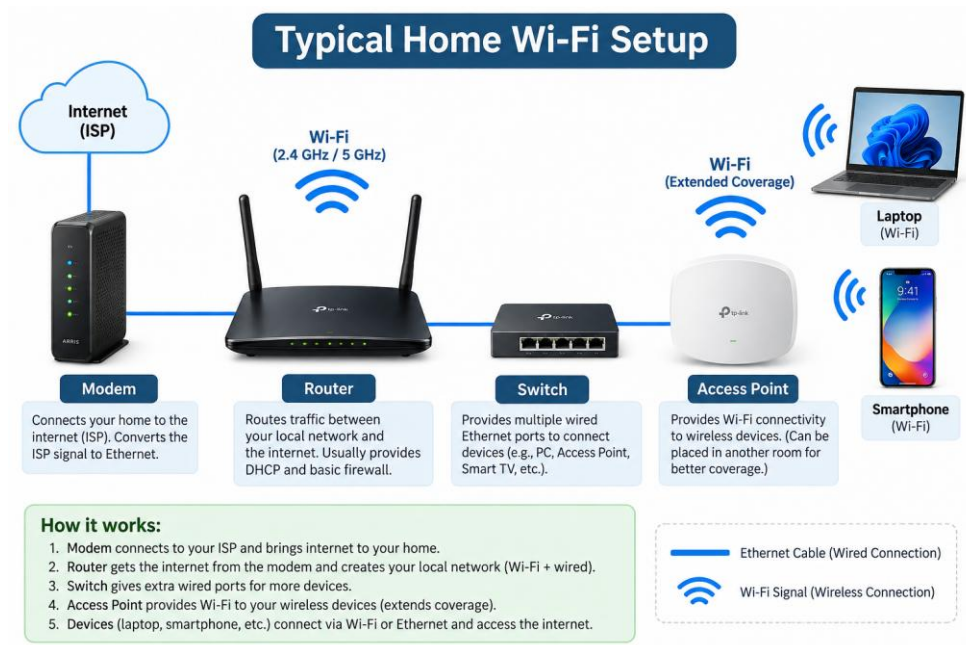


Day 1 Networking Hardware Theory

1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').

Networking Device	Main Function	Real-Life Example
Modem	Connects your home network to the ISP and converts the ISP signal into a usable network connection.	At home to receive internet from your ISP such as JioFiber, Airtel, or BSNL.
Router	Connects different networks and directs data between your local network and the internet. It also usually provides Wi-Fi and DHCP.	At home to share one internet connection with phones, laptops, TVs, etc.
Switch	Connects multiple wired devices within the same local network and forwards data to the correct device.	In an office to connect PCs, printers, servers, and other Ethernet devices.

Access Point (AP)	Provides wireless/Wi-Fi connectivity to devices and can extend Wi-Fi coverage.	In an office, school, or large house to provide Wi-Fi in areas where the main router's signal is weak.
Network Cable (Ethernet Cable)	Provides a physical wired connection for transmitting data between networking devices.	At home or in an office to connect a router to a switch, PC, TV, or access point.

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

1. Ethernet Cable (Cat5e/Cat6)



Common use: Used in homes and offices to connect computers, routers, switches, printers, and other network devices.

2. Coaxial Cable



Description : A cable with a central copper conductor surrounded by insulation and a protective metal shield.

Common use: Commonly used for cable TV, cable internet, satellite TV, and CCTV systems.

3. Fiber Optic Cable



Description : A cable containing thin glass or plastic fibers that transmit data as pulses of light.

Common Use: Used for high-speed internet, data centers, and long-distance telecommunications.

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Difference	Router	Switch
Main Use	Connects different networks and provides internet access to devices.	Connects multiple devices within the same local network (LAN).
Typical Location	Commonly found at home, offices, and network gateways.	Commonly found inside offices, server rooms, and data centers.
How It Handles Data	Forwards packets between different networks using IP addresses.	Forwards frames between devices using MAC addresses.
Network Layer	Primarily works at Layer 3 (Network layer) of the OSI model.	Primarily works at Layer 2 (Data Link layer) of the OSI model.
Internet Connection	Usually connects the local network to the internet/ISP.	Normally does not directly connect a LAN to the internet.
Example	A home Wi-Fi router shares one internet connection with phones, laptops, and TVs.	An office switch connects 20 PCs, printers, and other wired devices together.